

Homework 4

Question 1.

Find the gradients of the following functions without calculating a single partial derivative.

- (a) $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{x}) = \|\mathbf{x}\|.$$

Find the set of points where f is not differentiable.

- (b) $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{x}) = \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top B \mathbf{x}}$$

where $A, B \in \mathbb{S}_{++}^n$.

- (c) $f : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ defined by

$$f(X) = \det(X)$$

- (d) $f : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ defined by

$$f(X) = \frac{\det(X)}{\text{tr}(X)}$$

- (e) $f : \mathbb{R}^n \times \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{x}, Y) = \mathbf{x}^\top Y^{-1} \mathbf{x}$$

- (f) $f : \Omega \rightarrow \mathbb{R}$ defined by

$$\Omega = \{X \in \mathbb{R}^{m \times n} : X^\top A X + B^\top X + X^\top B + C \succ 0\}$$

with $A \in \mathbb{S}^m, B \in \mathbb{R}^{m \times n}, C \in \mathbb{S}^n$ arbitrary; and

$$f(X) = \log \det (X^\top A X + B^\top X + X^\top B + C)$$

Question 2.

Find the derivatives of the following functions without calculating a single partial derivative.

(a) $f : \mathbb{R}^4 \rightarrow \mathbb{R}^{2 \times 2}$ defined by

$$f(x, y, z, w) = \begin{bmatrix} x & y \\ z & w \end{bmatrix}^2 + \begin{bmatrix} x & y \\ z & w \end{bmatrix}^\top = X^2 + X^\top$$

(b) $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^n \setminus \{0\}$ defined by

$$f(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}$$

(c) $f : \mathbb{R}^{m \times n} \setminus \{0\} \rightarrow \mathbb{R}^{m \times n} \setminus \{0\}$ defined by

$$f(X) = \frac{X}{\|X\|}$$

(d) $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ defined by

$$f(X) = X^4$$

(e) $f : \text{GL}(n) \rightarrow \text{GL}(n)$ defined by

$$f(X) = X^{-1}$$

(f) $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ defined by

$$f(X) = \exp(X)$$

as in Problem 1 (a).